

FORTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY-
CONTROL ENGINEERING
MODEL QUESTION PAPER

Maximum marks 100

Time 3hours

PART A
(Maximum Marks:10)

- I Answer the following questions in one or two sentences. Each question carries 2 marks
- 1 Define control system.
 - 2 List any two advantages of signal flow graph over block diagram.
 - 3 Define type number.
 - 4 What is centroid in root locus?
 - 5 Prepare the first two rows of Rouths array for the characteristic equation $9S^5+S^4+2S+5=0$ (5x2=10)

PART B
(Maximum Marks:30)

- II Answer any five of the following questions. Each question carries 6 marks
- 1 Explain open loop control system with a suitable example.
 - 2 Define the following terms in one or two sentence
 - i) Node
 - ii) Transmittance
 - iii) Loop
 - 3 Explain any three standard test inputs with suitable graph
 - 4 Describe the steps to plot root locus
 - 5 Find the Laplace inverse of $F(s) = \frac{(s+5)}{(s+1)(s+2)}$
 - 6 Derive the transfer function of mechanical rotational system.
 - 7 Find an expression for steady state error of a type 1 system subjected to ramp input. (5x6=30)

PART -C

(Answer one full question from each module. Each question carries 15 marks.)

MODULE I

- III
- i State and prove real differentiation theorem. 6
 - ii Find the solution x(t) of the differential equation of the differential equation $\ddot{x} + 3\dot{x} + 2x = 0$ Given $x(0)=0, \dot{x}(0) = 1$ 9

OR

- IV
- i Find the Laplace transform of the function $f(t) = \cos at$. 6
 - ii Find the Laplace inverse of $F(s) = \frac{(s+2)}{(s+1)(s^2+6s+10)}$ 9

MODULE II

- V
- i State and Explain Masons Gain formula 6
 - ii Find the overall transfer function using block diagram reduction rules 9

OR

- VI
- i State any six block diagram reduction rules 6
 - ii Find the overall transmittance of the signal flow graph 9

MODULE III

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| VII | i | Define the following terms | 6 |
| | ii | Derive the time response of first order system for step input. | 9 |

OR

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|------|----|--|---|
| VIII | i | Define the following terms | 6 |
| | | i) Time constant ii) Delay time | |
| | | iii) Critical damping | |
| | ii | Derive the ramp response of a first order system | 9 |

MODULE IV

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| IX | ii | Comment on the stability of the system whose closed loop transfer function is given by $\frac{C(s)}{R(s)} = \frac{10}{3s^5 + 2s^3 + 2s^4 + 3s^2 + 5s^2 + 6s + 1}$ | 15 |
|----|----|---|----|

OR

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| X | i | Plot the root locus of open loop transfer function $K/(s)(s^2+9s+20)$ | 15 |
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